

# Venkata Kanna Bhavan Surya Adapa

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Software Engineer — Distributed Systems — Backend Development — Applied AI

## EDUCATION

**Amrita School of Engineering, Coimbatore**

B.Tech in Computer Science and Engineering

Aug 2023 – Aug 2027 (Expected)

CGPA: 8.4 / 10

## TECHNICAL SKILLS

**Core CS:** Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks, Distributed Systems

**Languages:** C++, Python, Java, C

**Backend:** FastAPI, Node.js, Express.js, REST APIs, WebSockets

**Frontend:** React, Flutter, Dart, HTML, CSS, JavaScript

**Databases:** PostgreSQL, MySQL, MongoDB, Redis

**Tools & Systems:** Docker, Git, JWT, Concurrent Programming, Linux

**Machine Learning:** PyTorch, scikit-learn, XGBoost, SHAP, Autoencoders

## PROJECTS

**ConversaVoice – Distributed Real-Time Voice AI Pipeline** — [GitHub](#)

- Architected a distributed voice AI system using asynchronous streaming and modular microservice design.
- Reduced response latency through Redis-based semantic caching and optimized request orchestration.
- Separated speech ingestion, reasoning, and synthesis pipelines for scalability and maintainability.
- Improved throughput and latency consistency under concurrent workloads through async task scheduling.

**RapidCover – Parametric Income Insurance Platform** — [GitHub](#)

- Built a full-stack insurance platform protecting gig workers against income loss caused by floods, heatwaves, AQI crises, and civic shutdowns.
- Developed a FastAPI backend with 18+ API modules supporting policy management, automated claims processing, and payout workflows.
- Integrated three ML models using XGBoost and Isolation Forest for risk scoring, premium prediction, and fraud detection.
- Engineered a fraud detection pipeline combining GPS validation, device fingerprinting, anomaly detection, and behavioral analytics achieving ROC-AUC of 0.995.
- Designed a zero-touch claim processing engine capable of validating disruption events and triggering payouts in under one minute.

**Hybrid Network Intrusion Detection System (ML + XAI)** — [GitHub](#)

- Developed a hybrid intrusion detection system using supervised classification and anomaly detection techniques.
- Implemented Autoencoder-based detection for identifying previously unseen cyber threats and zero-day attacks.
- Built a FastAPI-powered inference service enabling scalable deployment and experimentation.
- Achieved **99.56% accuracy** and **99.67% F1-score** on the CICIDS benchmark dataset.
- Applied SHAP-based explainability to improve model transparency and security analyst trust.

**Closer – Secure Real-Time Chat Application (MERN)** — [GitHub](#)

- Developed a real-time messaging platform using WebSockets for low-latency communication.
- Implemented JWT authentication, role-based access control, and secure session management.
- Designed RESTful APIs and scalable backend services using Node.js and Express.js.
- Enhanced communication security using RSA-2048 and AES-256-GCM hybrid encryption.

## CERTIFICATIONS & ACTIVITIES

**Tata – GenAI Powered Data Analytics Job Simulation (Forage)**

Dec 2025

- Completed hands-on simulation involving exploratory data analysis, AI-driven insights, and business problem solving.
- Worked with real-world datasets to derive actionable recommendations using analytical and GenAI techniques.

**NSS Volunteer**

2024 – 2025

**Hospitality Manager – Anokha 2024**